Vortical Enclosure Dynamics, Vol. 2

From Closure Geometry to the Standard Model

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Abstract

Vortical Enclosure Dynamics (VED) proposes formal constructions and candidate physical interpretations from a single axiom: *there is difference*. Vol. 1 established the causal log C_{ij} as the foundational variable and constructed a time-like accumulation quantity τ and network distance d_{ij}; their physical identifications remain proposed.

This volume extends the closure framework to the internal structure of the Standard Model. Beginning from the closure energy functional and the minimal condition for structural persistence, we propose structural comparisons in which the gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$, particle classification, generation structure, electric charge, the weak interaction, and the Higgs mechanism are related to closure geometry. These are model-level interpretations rather than observational identifications or universal derivations.

No additional structures are introduced beyond the closure framework. The present theory does not yet possess quantitative predictive power; structural organization is the primary goal.

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1 Introduction

Vol. 1 of this series introduced time-like and network-spatial constructions from the causal log C_{ij}, defined on a network of nodes subject to the single axiom *there is difference* [1]. The symmetric and antisymmetric decomposition

$$S_{ij} = \tfrac{1}{2}(C_{ij} + C_{ji}), \qquad A_{ij} = \tfrac{1}{2}(C_{ij} - C_{ji}) \tag{1}$$

encodes spatial and causal structure within the same variable.

The present volume addresses the next question: given those formal constructions and their proposed physical interpretations, can the internal structure of matter — particles, symmetries, charges, and masses — be understood within the same framework?

The Standard Model of particle physics describes this internal structure with remarkable precision. Its gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$, particle spectrum, charge assignments, generation structure, and mass-generation mechanism are experimentally confirmed, yet their geometric or structural origin remains unexplained.

The approach taken here is the following. Closure configurations are proposed as particle-like structural candidates, not identified with observed particles. Symmetries are identified not as a priori principles, but as transformation groups that leave closure invariant. Charges, generations, and mass receive model-level interpretations in the internal geometry of triangular closure.

The central claim of this volume is:

Closure geometry supplies a candidate structural comparison with Standard-Model organization.

(2)

This interpretation is structural and qualitative in nature. It does not constitute a predictive theory, and no claims are made about numerical coincidences with experimental values. The goal is to exhibit a coherent geometric reading of the Standard Model's skeleton.

The paper is organized as follows. Section 2 introduces the closure energy functional and defines model particle-like candidates as energy minima. Section 3 gives a model-local minimality argument for the triangle in a stated finite closure class. Section 4 derives the gauge symmetry structure from closure invariance. Sections 5–9 treat particle classification, generation structure, electric charge, the weak interaction, and the Higgs mechanism in turn. Section 10 discusses the relation to Vol. 1 and the path toward Vol. 3.

2 Closure Energy and Structural Foundations

2.1 From Causal Logs to Structure

Vol. 1 introduced the causal log C_{ij} and formal time-like and spatial constructions built from it:

$$\frac{d\tau_i}{dt} = \rho_i \qquad \text{(time)}, \tag{3}$$

$$d_{ij} = -\log \tilde{S}_{ij} \qquad \text{(space)}, \tag{4}$$

with the decomposition

$$S_{ij} = \tfrac{1}{2}(C_{ij} + C_{ji}), \qquad A_{ij} = \tfrac{1}{2}(C_{ij} - C_{ji}). \tag{5}$$

Spatial-candidate and causal structure are represented within the same object; physical identification is a separate question. The question addressed in this volume is not how structure is generated, but:

$$\boxed{\text{What configurations of this structure can persist?}} \tag{6}$$

2.2 Minimal Condition for Structural Persistence

Consider the simplest configurations. Two nodes define only a relation (a line); three nodes allow a loop. A loop is the first candidate in the finite simple-loop class used here under which a structure can reference itself and maintain internal consistency.

$$\boxed{\text{Closure first becomes possible at three nodes.}} \tag{7}$$

Two-node systems do not enclose an area in this model class. Three-node systems introduce the first candidate enclosure; persistence is not inferred without a separate dynamical analysis.

2.3 Internal Representation from Logs

Each node i carries an effective internal vector $\vec{e}_i$, defined entirely from C_{ij}:

$$\langle \vec{e}_i, \vec{e}_k \rangle = \sum_{j,\ell} A_{ij} A_{k\ell} \cdot S_{ij} S_{k\ell} \cdot \tilde{S}_{j\ell}. \tag{8}$$

No external embedding space is assumed. Direction, distance, and relative orientation arise from internal consistency conditions of the log network.

$$\boxed{\text{Geometry is induced from relational structure, not presupposed.}} \tag{9}$$

2.4 Triangular Closure Measure

For a triplet (i, j, k), define the triangular closure strength:

$$\phi_{ijk} = \exp\Big(-\eta \left|\vec{e}_i + \vec{e}_j + \vec{e}_k\right|^2\Big) \cdot \frac{1}{2} \left|(\vec{e}_i - \vec{e}_k) \times (\vec{e}_j - \vec{e}_k)\right|. \tag{10}$$

The first factor measures closure consistency (how close the vector sum is to zero); the second measures non-degeneracy (whether the triangle has area). Total closure strength over a set A:

$$\Phi(A) = \sum_{i<j<k \in A} \phi_{ijk}. \tag{11}$$

For $n = 2$, $\Phi = 0$; area and closure emerge first at $n = 3$.

2.5 Closure Energy Functional

To determine which structures persist, define the closure energy:

$$\boxed{E(A) = \alpha \left|\sum_{i \in A} \vec{e}_i\right|^2 - \beta\, \Phi(A) + \gamma |A| + B(L(A))} \tag{12}$$

The terms have the following roles:

Term	Interpretation
$\alpha \lvert \sum \vec{e}_i \rvert^2$	penalty for incomplete closure
$-\beta\, \Phi(A)$	reward for triangular closure
$\gamma \lvert A \rvert$	structural cost
$B(L(A))$	effective non-closure resistance near the complete-closure limit

The final term is a phenomenological encoding of the non-closure constraint inherited from Vol. 1.

Equation (12) is the model energy / scoring functional used to rank configurations within this closure model. Its mathematical role is distinct from the subsequent physical interpretation of a local minimum.

2.6 Particles as Stable Configurations

Structures evolve under this energy functional. The model associates candidate configurations with local minima:

$$\boxed{\text{A model-defined particle is a local minimum of } E(A).} \tag{13}$$

Such minima are particle-like structural candidates rather than established identifications with observed particles.

$$\boxed{\text{Closure-energy minima are candidate particle-like structures.}} \tag{14}$$

3 Minimal Closure and Triangular Geometry

3.1 Reformulating the Question

Section 2 introduced closure as the condition for structural persistence, and identified three-node configurations as the first candidates for such structures. The central question is:

$$\boxed{\text{Why is the minimal closure unit triangular?}} \tag{15}$$

The following finite argument tests this question within the stated simple-loop closure class; it does not establish unrestricted topological necessity.

3.2 Two-Node Systems: Absence of Closure

A system of two nodes permits only a bidirectional relation $i \leftrightarrow j$. Such a system defines a connection, but not an enclosure: no loop is formed, no area is defined, and no internal reference is possible. Accordingly, $\Phi = 0$.

$$\boxed{\text{A two-node system cannot form a self-sustaining structure.}} \tag{16}$$

3.3 Three-Node Systems: Emergence of Closure

With three nodes (i, j, k), a closed loop $i \to j \to k \to i$ becomes possible. This configuration introduces a minimal loop, a definable area, and mutual reference among nodes. The closure condition is:

$$\vec{e}_i + \vec{e}_j + \vec{e}_k \approx 0, \tag{17}$$

which encodes vector balance and structural consistency and is used as a candidate enclosure condition; stability requires separate dynamics.

$$\boxed{\text{Three nodes form the minimal self-referential enclosure.}} \tag{18}$$

3.4 Higher-Node Systems: Decomposition into Triangles

For $n \geq 4$, the finite configurations considered here are represented by triangular components:

$$\text{closure}(n) \longrightarrow \sum \text{triangular closures.} \tag{19}$$

No additional minimal unit appears within this stated construction class.

$$\boxed{\text{Triangles are the minimal units in the stated closure class.}} \tag{20}$$

3.5 Uniqueness of the Triangle

Node count	Structural property
2	No closure
3	Minimal closure
4+	Composite of triangles

$$\boxed{\text{The triangle is minimal in the stated finite closure class.}} \tag{21}$$

This model-local minimality follows from the listed admissible configurations. It does not assert uniqueness over every possible definition of closure or equivalence of higher-node configurations.

3.6 Modes of Triangular Closure

Not all triangular closures are equivalent. Three structural regimes can be identified:

1. **Non-degenerate triangle** — finite area, three independent directions, maximal structural stability.

2. **Degenerate triangle** — area approaches zero, reduced dimensionality, constrained degrees of freedom.

3. **Extreme degeneracy** — collapse of area, effectively point-like, minimal internal structure.

$$\boxed{\text{Structural behavior depends on the degree of triangular degeneracy.}} \tag{22}$$

These three regimes are summarized in Fig. 1.

3.7 Toward Symmetry

Within the stated finite closure class, the triangle is the minimal candidate, but it also carries internal degrees of freedom. The next step is to understand which

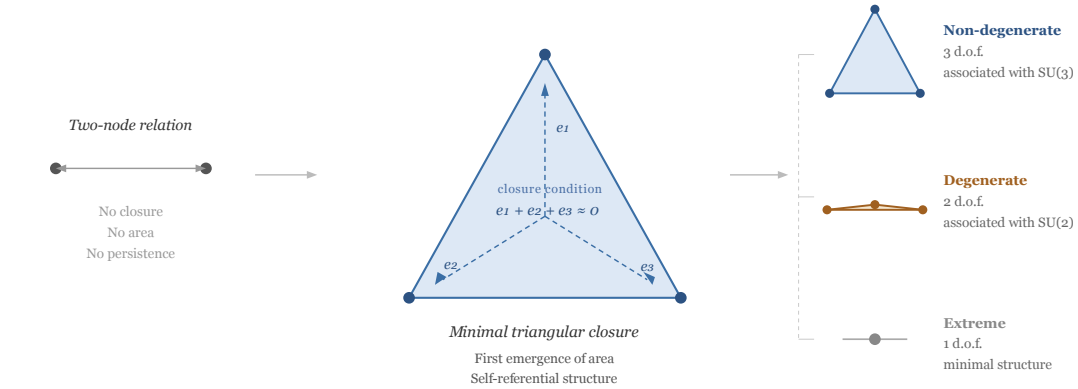


Figure 1: Closure does not occur in two-node systems, where no loop or area is defined. The triangle is the minimal candidate configuration in the stated class, where vector balance and enclosed area first emerge. Triangular closure admits three structural regimes depending on degeneracy: non-degenerate (full area), degenerate (collapsed area), and extreme degeneracy. These regimes correspond to different structural degrees of freedom, which are associated with distinct symmetry structures in subsequent analysis.

transformations leave closure invariant. This leads directly to the emergence of gauge symmetry in the following section.

4 Emergence of Symmetry from Closure Geometry

4.1 Reversing the Perspective

In conventional formulations of particle physics, gauge symmetries such as SU(3), SU(2), and U(1) are introduced as fundamental principles. Within the present framework, the perspective is reversed. Since particles are interpreted as stable configurations of triangular closure, the relevant question becomes:

$$\boxed{\text{What transformations preserve closure structure?}} \tag{23}$$

Symmetry is not assumed a priori; it is identified through invariance of structure.

4.2 Definition of Symmetry

A symmetry is defined operationally as:

$$\boxed{\text{A transformation that preserves closure conditions.}} \tag{24}$$

The closure condition consists of vector balance, internal consistency, and non-degeneracy (when applicable). Admissible transformations must leave invariant $\vec{e}_i + \vec{e}_j + \vec{e}_k \approx 0$ and the associated relational structure.

4.3 Non-Degenerate Triangle and SU(3)

A non-degenerate triangular closure has three internal components subject to one constraint:

$$\vec{e}_1 + \vec{e}_2 + \vec{e}_3 = 0. \tag{25}$$

Transformations acting on these internal components must preserve the closure condition and inner products (interaction structure). This leads naturally to unitary transformations:

$$\psi \mapsto U\psi, \qquad U^\dagger U = \mathbf{1}. \tag{26}$$

Fixing the overall scale ($\det U = 1$) yields:

SU(3) can be identified as a minimal transformation group consistent with the invariance of non-degenerate triangular closure.	(27)

4.4 Degenerate Triangle and SU(2)

When the triangle degenerates, one directional degree of freedom is effectively lost. The structure reduces to two distinguishable internal states. Transformations preserving this reduced closure structure act on a two-dimensional internal space:

$$\begin{pmatrix} |L\rangle \\ |R\rangle \end{pmatrix} \longrightarrow U \begin{pmatrix} |L\rangle \\ |R\rangle \end{pmatrix}, \qquad U \in \mathrm{SU}(2). \tag{28}$$

SU(2) is associated with the internal degrees of freedom of degenerate triangular closure.	(29)

4.5 Global Phase and U(1)

There exists a transformation that uniformly rescales all internal vectors by a phase:

$$\vec{e}_i \;\to\; e^{i\theta}\vec{e}_i. \tag{30}$$

This transformation preserves the closure condition and does not alter relative structure; it represents a redundancy rather than a physical change.

$$\boxed{\text{U(1) is a global phase symmetry that leaves closure structure invariant.}} \tag{31}$$

4.6 Structural Correspondence

The correspondence between closure structure and transformation groups is summarized in Table 1 and Fig. 2.

Closure structure	Associated symmetry
Non-degenerate triangle	SU(3)
Degenerate triangle	SU(2)
Global phase redundancy	U(1)

Table 1: Correspondence between closure regimes and symmetry groups.

$$\boxed{\text{SU(3)} \times \text{SU(2)} \times \text{U(1)} \text{ emerges as the symmetry structure of closure geometry.}} \tag{32}$$

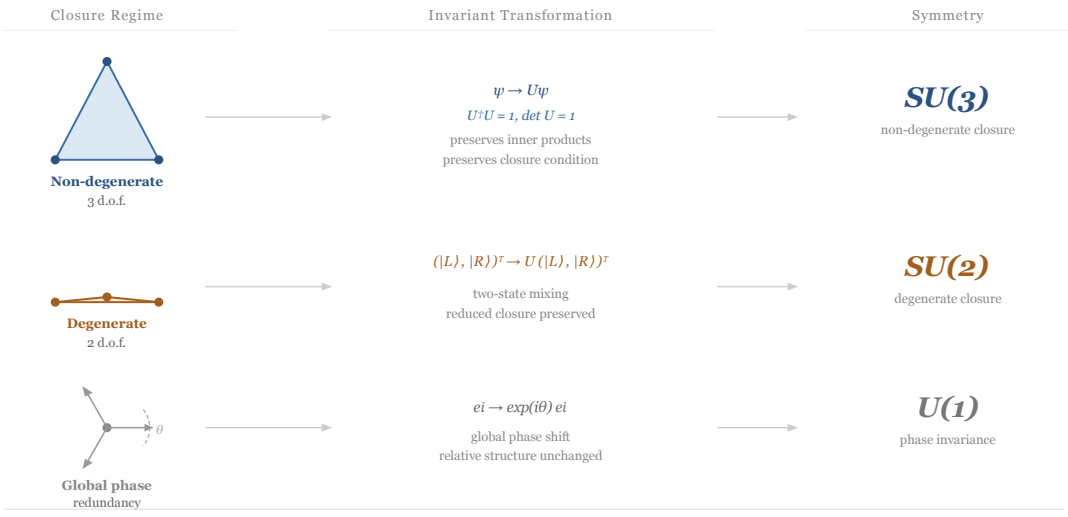


Figure 2: Symmetry structure as invariance of closure geometry. Distinct closure regimes admit distinct classes of transformations that preserve their internal structure. Non-degenerate triangular closure is associated with SU(3), degenerate closure with SU(2), and global phase redundancy with U(1). Symmetry is interpreted not as a starting assumption, but as the class of admissible transformations preserving closure.

4.7 The Inversion

The conventional logical order is: symmetry $\rightarrow$ particle. The present framework reverses this:

$$\boxed{\text{closure structure } \rightarrow \text{ allowed transformations (symmetry).}} \tag{33}$$

Symmetry is therefore not a starting assumption, but a reflection of structural invariance.

5 Classification of Particles as Closure Structures

5.1 Reformulating Particle Identity

In conventional particle physics, particles are treated as fundamental entities belonging to distinct categories. Within the present framework, closure configurations are proposed as particle-like structural candidates:

$$\boxed{\text{Could observed particle classes admit a closure-structure map?}} \tag{34}$$

The model classification is therefore organized by closure geometry. An observational identification is not supplied here.

5.2 Degeneracy as the Organizing Principle

Section 3 established that triangular closure admits different structural regimes depending on degeneracy. This provides a natural organizing principle:

$$\boxed{\text{Model particle-like classes are organized by triangular degeneracy.}} \tag{35}$$

Degeneracy controls the number of effective degrees of freedom, the symmetry structure, and the stability characteristics of each configuration.

5.3 Three Structural Regimes

Non-degenerate triangular closure. Finite area, three independent directions, maximal internal coupling.

$$\boxed{\begin{array}{c}\text{Non-degenerate closure represents the maximally articulated} \\ \text{configuration within the triangular regime.}\end{array}} \tag{36}$$

Degenerate triangular closure. Area approaches zero, one direction collapses, two effective degrees of freedom remain.

$$\boxed{\text{Degenerate closure represents a reduced structural configuration.}} \tag{37}$$

Extreme degeneracy. Area effectively vanishes, structure collapses toward a point, minimal internal interaction.

$$\boxed{\text{Extreme degeneracy corresponds to minimal closure structure.}} \tag{38}$$

5.4 Structural Interpretation of Particle Types

These three regimes are compared with familiar particle classes, as summarized in Table 2 and Fig. 3.

Closure regime	Structural interpretation
Non-degenerate triangle	baryon-like structures
Degenerate triangle	lepton-like structures
Extreme degeneracy	neutrino-like structures

Table 2: Proposed structural correspondence between closure regimes and particle-like classes. No observational identification is asserted.

$$\boxed{\text{Closure regimes provide candidate particle-like classes.}} \tag{39}$$

5.5 The Degeneracy Ladder

The three regimes form a continuous hierarchy:

$$\text{non-degenerate} \;\rightarrow\; \text{degenerate} \;\rightarrow\; \text{extreme}. \tag{40}$$

$$\boxed{\text{Candidate particle-like structures form a degeneracy hierarchy.}} \tag{41}$$

Candidate discrete types can be represented as points along this continuous structure; stability and observational mapping remain open.

5.6 Mass as Distance from Closure

Define the closure deficit relative to the complete-closure limit:

$$\Delta_k = L_{\max} - L_k. \tag{42}$$

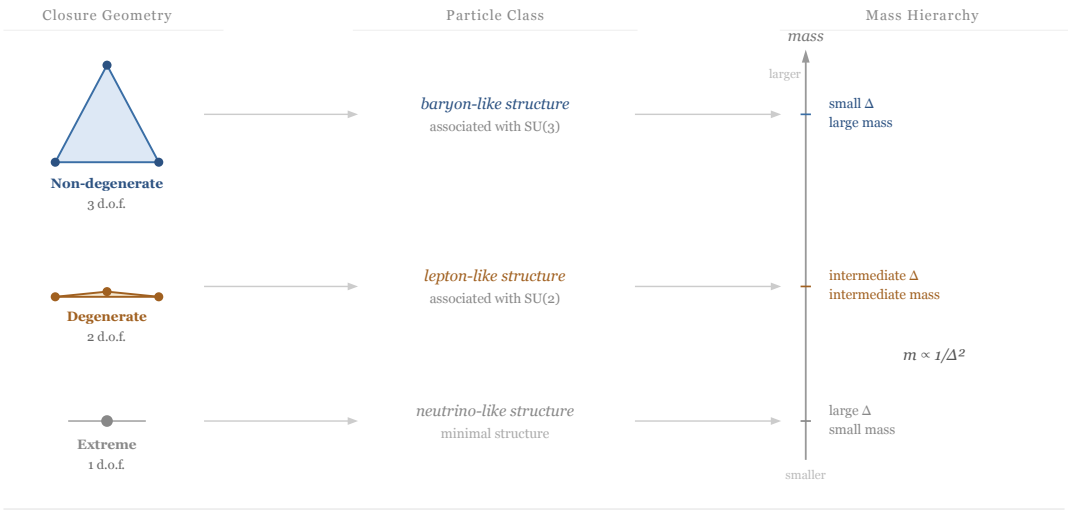


Figure 3: Particle classes as closure geometry. Triangular closure admits three structural regimes depending on degeneracy. These regimes are compared with baryon-like, lepton-like, and neutrino-like structures. A mass-like scale is associated phenomenologically with distance from the complete-closure limit, with $m \propto 1/\Delta^2$.

Mass scales as:

$$m \propto \frac{1}{\Delta_k^2}. \tag{43}$$

This is a phenomenological barrier-based scaling within the closure model; it does not by itself derive or calibrate an observed particle mass. Stronger closure implies smaller Δ and larger mass; weaker closure implies larger Δ and smaller mass.

$$\boxed{\text{A mass-like scale is modeled by closure-limit distance.}} \tag{44}$$

Here the complete-closure limit is not assumed to be an attainable physical state. It is a limiting representation of the point at which the internal differences required for persistent structure would disappear.

5.7 Discrete Particles from Continuous Geometry

Although closure geometry varies continuously, the model asks how discrete particle-like candidates might be represented:

$$\boxed{\begin{array}{c}\text{Discrete particle-like candidates may be represented by points} \\ \text{in a continuous geometric landscape.}\end{array}} \tag{45}$$

This is a proposed structural classification, not an established map to observed particle species.

6 Generation Structure as Sequential Closure

6.1 The Question of Generations

The Standard Model contains three generations of fermions. While this structure is experimentally established, its origin is not explained within the conventional framework. Within the present approach, the question is reformulated:

$$\boxed{\text{Can the number of generations be understood from closure geometry?}} \tag{46}$$

6.2 Structural Basis: The Triangle

The minimal closure unit identified in Section 3 is the triangle, consisting of three nodes and three edges. Closure is not a single event, but a process involving relations along these edges.

$$\boxed{\text{Closure develops through interactions along the edges of the triangle.}} \tag{47}$$

6.3 Sequential Stabilization

Closure need not occur simultaneously across all edges. Instead, stabilization may proceed progressively:

Stabilized edges	Structural state
1	partially stabilized
2	intermediate stabilization
3	fully stabilized

$$\boxed{\text{Closure can be interpreted as a sequential stabilization process.}} \tag{48}$$

6.4 Interpretation as Generational Structure

This sequential stabilization suggests a correspondence between stabilization stage and generation:

Stabilization stage	Interpretation
1 edge stabilized	first generation (lighter)
2 edges stabilized	second generation (intermediate)
3 edges stabilized	third generation (heavier)

$$\boxed{\text{Generations can be interpreted as stages of closure stabilization.}} \tag{49}$$

This relates generational hierarchy to structural progression rather than independent particle families.

6.5 Why Three

The number of stages is determined by the number of edges in the minimal closure unit. Since a triangle has exactly three edges:

$$\boxed{\text{The number of generational stages is bounded by three.}} \tag{50}$$

No additional independent stage is available within the same minimal structure.

6.6 Non-Simultaneity and Asymmetry

Simultaneous stabilization of all edges would correspond to the limiting case of a perfectly symmetric configuration. However, from the foundational principle established in Vol. 1:

$$\boxed{\text{Perfect symmetry does not produce dynamics.}} \tag{51}$$

Stabilization must therefore proceed asymmetrically, with edges locking at distinct stages.

$$\boxed{\text{Sequential stabilization is structurally favored over simultaneous closure.}} \tag{52}$$

6.7 Relation to Mass Hierarchy

As stabilization progresses, closure strength increases and the closure deficit Δ decreases. Given equation (43):

- early-stage structures $\rightarrow$ larger $\Delta \rightarrow$ lower mass,
- later-stage structures $\rightarrow$ smaller $\Delta \rightarrow$ higher mass.

$$\boxed{\text{Sequential closure motivates a phenomenological mass-like hierarchy.}} \tag{53}$$

This is a model-local use of the mass-like scaling ansatz; it does not derive observed generation masses.

6.8 Structural Interpretation

$$\boxed{\begin{array}{c}\text{Generations are not distinct particle types, but different} \\ \text{stability stages of the same underlying structure.}\end{array}} \tag{54}$$

This contrasts with the conventional view in which generations are treated as independent copies.

7 Electric Charge as Phase Asymmetry in Closure

7.1 Reformulating the Origin of Charge

In the Standard Model, electric charge is assigned as a fundamental property of particles. Within the present framework, charge is examined as a structural feature of triangular closure:

$$\boxed{\text{Can electric charge be understood as an internal asymmetry of closure structure?}} \tag{55}$$

7.2 Symmetric Phase Configuration

Consider a non-degenerate triangular closure. Let each node carry a phase-like quantity ϕ_i, normalized such that:

$$\sum_{i=1}^{3} \phi_i = 1. \tag{56}$$

The symmetric configuration is:

$$\phi_1 = \phi_2 = \phi_3 = \frac{1}{3}. \tag{57}$$

This configuration is structurally balanced — no node is preferred, no directional bias exists.

$$\boxed{\text{The symmetric configuration corresponds to zero net charge.}} \tag{58}$$

7.3 Introducing Asymmetry

When the phase distribution becomes asymmetric, deviations from the mean arise. Define charge-like quantities as:

$$q_i = \phi_i - \frac{1}{3}. \tag{59}$$

For example, a maximally asymmetric configuration $(\phi_1, \phi_2, \phi_3) = (1, 0, 0)$ yields:

$$q = \left(+\frac{2}{3},\ -\frac{1}{3},\ -\frac{1}{3} \right). \tag{60}$$

$$\boxed{\text{Fractional values such as } +\tfrac{2}{3} \text{ and } -\tfrac{1}{3} \text{ emerge naturally from three-node asymmetry.}} \tag{61}$$

7.4 Charge as Deviation from Symmetry

Charge can be interpreted as deviation from the symmetric phase configuration. (62)

This implies that charge is relational rather than intrinsic — it reflects internal imbalance within closure.

7.5 Charge Conservation

By construction:

$$\sum_i q_i = 0. \tag{63}$$

Charge conservation follows from normalization of the internal structure. (64)

No additional conservation law needs to be imposed.

7.6 Fractional Structure

The appearance of fractional values is a direct consequence of the three-node structure. The natural unit is 1/3.

Fractional charges can be understood as a geometric consequence of triangular closure. (65)

7.7 Degenerate Closure and Integer Charges

In the degenerate regime, effective dimensionality reduces and the three-node structure collapses toward a two-state system. The phase structure simplifies, and integer-like charge values emerge.

Integer charges can be associated with reduced (degenerate) closure structures. (66)

In the extreme degenerate limit, $q \approx 0$, corresponding to minimal interaction.

7.8 Structural Interpretation

Charge is not an assigned quantity, but a manifestation of internal asymmetry in closure geometry. (67)

This provides a structural basis for both fractional charge values and charge conservation.

8 Weak Interaction as Structural Fluctuation of Degenerate Closure

8.1 Reformulating the Weak Interaction

In the Standard Model, the weak interaction is described by an SU(2) gauge symmetry and couples only to left-handed fermions. Within the conventional framework, both the origin of SU(2) and the left-handed asymmetry are postulated rather than derived. Within the present approach, the question is reformulated:

$$\boxed{\text{Can the weak interaction be understood as an internal structural process?}} \tag{68}$$

8.2 Degenerate Triangular Closure

Section 5 identified degenerate triangular closure as a reduced structural regime. In this limit, the area approaches zero, one degree of freedom collapses, and two effective internal states remain:

$$\boxed{|L\rangle \quad \text{and} \quad |R\rangle.} \tag{69}$$

This two-state structure forms the basis for internal transitions.

8.3 Reduction of Degrees of Freedom

The transition from non-degenerate to degenerate closure reduces the number of effective internal directions:

$$\mathrm{SU}(3) \longrightarrow \mathrm{SU}(2). \tag{70}$$

$$\boxed{\text{Degeneracy induces a reduction of symmetry from SU(3) to SU(2).}} \tag{71}$$

8.4 Internal Mixing as Structural Fluctuation

The two-state system is not static. It admits internal transformations that preserve reduced closure structure while mixing the internal states:

$$\begin{pmatrix} |L\rangle \\ |R\rangle \end{pmatrix} \longrightarrow U \begin{pmatrix} |L\rangle \\ |R\rangle \end{pmatrix}, \qquad U \in \mathrm{SU}(2). \tag{72}$$

$$\boxed{\text{Weak interaction can be interpreted as mixing within degenerate closure.}} \tag{73}$$

8.5 Origin of Chirality

A key feature of the weak interaction is its preference for left-handed states. Within this framework, perfect degeneracy would imply symmetry between $|L\rangle$ and $|R\rangle$. However,

from the foundational principle established in Vol. 1:

$$\boxed{\text{Perfect symmetry is dynamically inaccessible.}} \tag{74}$$

Residual asymmetry is unavoidable. One state becomes structurally favored; the other is suppressed.

$$\boxed{\begin{array}{c}\text{Chirality can be interpreted as a consequence of residual} \\ \text{asymmetry in degenerate closure.}\end{array}} \tag{75}$$

8.6 Structural Interpretation of Decay

Weak processes appear as particle decay. Within the present framework, such processes correspond to structural reorganization:

$$|L\rangle \;\leftrightarrow\; |R\rangle. \tag{76}$$

$$\boxed{\text{Decay can be interpreted as reconfiguration of closure structure.}} \tag{77}$$

This shifts the description from force-mediated interaction to structural transformation.

8.7 Relation to CP Asymmetry

Since perfect symmetry is inaccessible, residual imbalance may also manifest as asymmetry under combined transformations.

$$\boxed{\begin{array}{c}\text{CP asymmetry can be interpreted as a structural consequence} \\ \text{of incomplete degeneracy.}\end{array}} \tag{78}$$

This suggests that symmetry violations reflect underlying geometric constraints, rather than constituting independent phenomena.

8.8 Structural Interpretation

$$\boxed{\begin{array}{c}\text{Weak interaction is not introduced as a force, but interpreted} \\ \text{as fluctuation within a degenerate closure structure.}\end{array}} \tag{79}$$

This provides a unified structural basis for SU(2) symmetry, state mixing, chirality, and decay processes.

9 Structural Fixing and the Higgs Interpretation

9.1 Reformulating the Higgs Mechanism

In the Standard Model, the Higgs mechanism is introduced to explain symmetry breaking and mass generation. The Higgs field is treated as an additional entity imported for this purpose. Within the present framework, the question is reformulated:

$$\boxed{\text{Can symmetry breaking and mass arise from internal structural dynamics?}} \tag{80}$$

9.2 Fluctuating Degenerate Structure

Section 8 showed that degenerate triangular closure admits internal fluctuation between two states:

$$|L\rangle \;\leftrightarrow\; |R\rangle. \tag{81}$$

In this regime, no configuration is uniquely preferred and the structure remains dynamically unstable.

$$\boxed{\text{The degenerate closure is intrinsically undecided.}} \tag{82}$$

9.3 Order Parameter for Structural Asymmetry

To characterize this instability, introduce an order parameter:

$$\varphi = |\psi_L|^2 - |\psi_R|^2. \tag{83}$$

This quantity measures the imbalance between the two internal states.

$$\boxed{\varphi \text{ quantifies structural asymmetry in degenerate closure.}} \tag{84}$$

9.4 Fixing of Asymmetry

Stability is achieved when the system selects a preferred configuration, $\varphi \neq 0$. Fluctuation is suppressed, a dominant state emerges, and the structure stabilizes.

$$\boxed{\text{Fixing of asymmetry leads to stable closure.}} \tag{85}$$

This process corresponds to what is conventionally described as spontaneous symmetry breaking.

9.5 Relation to Mass

As asymmetry becomes fixed, closure becomes more stable and the closure deficit Δ decreases. Given equation (43):

$$\boxed{\text{Stabilized asymmetry motivates a phenomenological mass-like scale.}} \tag{86}$$

9.6 Effective Potential

The dynamics of the order parameter are described by an effective potential:

$$V(\varphi) = a\varphi^2 + b\varphi^4 - h\varphi. \tag{87}$$

For $a < 0$, a nonzero minimum emerges:

$$\langle\varphi\rangle = v_\varphi \neq 0. \tag{88}$$

$$\boxed{\text{A nonzero expectation value corresponds to structural fixing.}} \tag{89}$$

The linear term h reflects residual asymmetry intrinsic to the closure structure — specifically, the imbalance between left and right edges and the causal flow asymmetry of the degenerate triangle.

9.7 Structural Interpretation of the Higgs Field

Within this framework, the Higgs is not introduced as an external field, but is associated with an internal degree of freedom already present in the degenerate closure.

$$\boxed{\text{The Higgs can be interpreted as the order parameter governing structural asymmetry.}} \tag{90}$$

9.8 Vacuum Expectation Value

The vacuum expectation value v_φ characterizes the stable configuration:

$$v_\varphi \sim \sqrt{\frac{\alpha_1 U(A) + \alpha_2\, \delta_{\text{asym}}(A) - a_0}{2\kappa_B}}. \tag{91}$$

This expression encodes non-closure contributions, asymmetry contributions, and structural constraints.

$$\boxed{\text{The vacuum state corresponds to a stabilized closure configuration.}} \tag{92}$$

9.9 Structural Inversion

The conventional interpretation is: symmetry breaking $\rightarrow$ mass. Within this framework:

$$\boxed{\text{structural fixing } \rightarrow \text{ phenomenological mass-like scale.}} \tag{93}$$

Within this phenomenological model, a mass-like scale is associated with stabilization of internal structure; observed mass is not derived here.

9.10 Summary of Vol. 2

The structural correspondence developed across this volume:

$$\boxed{\begin{aligned} \text{Particle-like candidate} &= \text{triangular closure} \\ \text{Symmetry} &= \text{invariance of closure} \\ \text{Generation} &= \text{sequential stabilization} \\ \text{Charge} &= \text{phase asymmetry} \\ \text{Weak interaction} &= \text{structural fluctuation} \\ \text{Mass-like scale} &= \text{stabilized asymmetry} \end{aligned}} \tag{94}$$

10 Discussion

10.1 Relation to Vol. 1

Vol. 1 constructed time-like and spatial quantities from the causal-log model; their physical identifications remain open. The present volume proposes that the internal structure of matter — particle types, symmetries, charges, and masses — can be read from the same closure geometry.

The key bridge is the closure energy functional (12). In Vol. 1, this functional was used in a candidate metric interpretation. In the present volume, its local minima are compared with particle-like types, and its symmetry classes with gauge-group structure. The two volumes thus share a common geometric foundation.

10.2 Scope and Limitations

The interpretations advanced in this volume are structural and qualitative. Several points of caution are warranted.

First, no quantitative predictions are derived. The mass relation $m \propto 1/\Delta^2$ and the charge formula $q_i = \phi_i - 1/3$ are structural analogies, not derived equalities. Precise numerical correspondence with observed values would require a more developed formalism.

Second, the identification of closure regimes with particle classes is proposed as a consistent mapping, not as a proof of equivalence. The claim is that non-degenerate triangular closure *admits* the structure of baryonic states, not that it *is* a baryon.

Third, the derivation of SU(3) from non-degenerate closure identifies it as a *candidate* consistent with closure invariance, not as a unique derivation. A rigorous uniqueness argument would require additional constraints.

10.3 Path toward Vol. 3

Vol. 3 addresses candidate large-scale interpretations of closure geometry, including gravity-like and spacetime-curvature comparisons based on closure density, vorticity, and metric distortion. The central question is whether the Einstein equations can be read as a fixed-point condition of the closure network.

The present volume provides the particle-physics half of this picture. The two halves — internal structure (Vol. 2) and large-scale geometry (Vol. 3) — are compared within the same closure framework; no exact unification is established by that shared setting.

11 Conclusion

This volume has shown that the structure of the Standard Model can be interpreted as arising from closure geometry. Beginning from the closure energy functional and the minimal condition for structural persistence, we have identified:

- the triangle as the minimal candidate in the stated finite closure class (Section 3);
- $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ as the symmetry structure of closure invariance (Section 4);
- particle-like candidate classes as structural regimes of triangular degeneracy (Section 5);
- generations as sequential stabilization of triangular edges (Section 6);
- electric charge as phase asymmetry within closure (Section 7);
- the weak interaction as internal fluctuation of degenerate closure (Section 8);
- a phenomenological mass-like and Higgs comparison based on structural fixing of asymmetry (Section 9).

The following compact display records the program's navigation order:

$$\text{difference} \;\rightarrow\; C_{ij} \;\rightarrow\; \text{triangular closure} \;\rightarrow\; \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1) \;\rightarrow\; \text{Standard Model}. \tag{95}$$

Its arrows do not share one logical status. The passage from C_{ij} to a triangle is a model-local finite closure construction in the stated admissible class; the gauge-group step is a proposed symmetry-structure comparison; and the Standard Model is a physical comparison target rather than a derived endpoint. The display therefore does not establish a universal generation mechanism or an identification with the observed theory.

The Standard Model is used as a comparison target for a proposed structural interpretation based on closure geometry. This interpretation is qualitative in nature and does not yet constitute a predictive theory. The path toward quantitative consequences and large-scale geometry is taken up in Vol. 3.

References

[1] Yuhki Endoh. Vortical enclosure dynamics, vol. 1: From difference to space and time, 2026. Independent research draft.

A Notation

Symbol	Definition
C_{ij}	causal log from node j to node i
S_{ij}	symmetric part of C_{ij}: $\frac{1}{2}(C_{ij} + C_{ji})$
A_{ij}	antisymmetric part of C_{ij}: $\frac{1}{2}(C_{ij} - C_{ji})$
$\vec{e}_i$	effective flux vector of node i
ϕ_{ijk}	non-degeneracy measure of triangle (i, j, k)
$\Phi(A)$	total closure strength over set A
$E(A)$	closure energy
$\Delta(A)$	degeneracy measure (area element of triangle)
Δ_k	mass barrier distance $= L_{\max} - L_k$
ℓ_{ab}	effective edge coupling
L_k	lock threshold for edge k
$\mathcal{R}(A)$	closure ratio (internal/external coupling ratio)
$U(A)$	non-closure measure $= 1/(1 + \mathcal{R}(A))$
ϕ_i	phase-like quantity at node i; $\sum_i \phi_i = 1$
q_i	charge-like quantity $= \phi_i - 1/3$
φ	Higgs order parameter $= \lvert\psi_L\rvert^2 - \lvert\psi_R\rvert^2$
v_φ	vacuum expectation value of φ
θ_W	Weinberg angle
$\mathcal{A}_{ij}$	U(1) connection (electromagnetic potential)
F_{ijk}	electromagnetic field strength (curvature)